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Validation of Bioelectrical Impedance Devices for the Determination of Body Fat Percentage in Firefighters

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Abstract

Jagim, AR, Luedke, J, Erickson, JL, Fields, JB, and Jones, MT. Validation of bioelectrical impedance devices for the determination of body fat percentage in firefighters. *J Strength Cond Res* 38(8): e448–e453, 2024—To cross-validate bioelectrical impedance devices for the determination of body fat percentage (BF%) in firefighters. Twenty-eight structural firefighters were evaluated (female, $n = 2$; male, $n = 26$ [mean ± *SD*] age: 38.2 ± 8.3 years; height: 180.2 ± 7.5 cm; body mass: 86.7 ± 20.8 kg; body mass index: 25.8 ± 7.8 kg·m^{-2}) using multifrequency bioelectrical impedance analysis (MFBIA) hand-to-foot device, and single-frequency BIA foot scale (F2FBIA), and a single-frequency handheld BIA device (HHBIA). Dual X-ray absorptiometry served as the criterion. Validity metrics were examined to establish each method's performance. Body fat % values produced by MFBIA ($r = 0.913$), F2FBIA ($r = 0.695$), and HHBIA ($r = 0.876$) were strongly associated ($p < 0.001$) with criterion BF% measures. However, MFBIA, F2FBIA, and HHBIA all significantly ($p < 0.001$) underestimated BF% when compared with the criterion measure. Constant error ranged between 4.0 and 5.5% across all BIA devices. Despite strong associations between the BIA devices included in the current study and the criterion measure, all BIA devices underestimated BF%, which resulted in an overestimation of fat-free mass. In addition, proportional bias was observed in which BF% was overestimated at lower values and underestimated at higher values.

Key Words: body composition, bioelectrical impedance analysis, tactical athletes, fat-free mass, first responders

Introduction

Structural firefighting is a physically demanding occupation that requires a high degree of aerobic and anaerobic fitness, muscular strength, endurance, explosive power, and reaction time (28). Structural firefighters are on duty year-round and respond to emergency calls in urban communities as opposed to wildland firefighters who are often seasonal and work in rural areas. Although fire suppression activities may present acute health hazards, nearly 50% of all on-duty deaths occur from cardiac events (6). It has been postulated that health risks incurred from hazardous environment conditions, disrupted sleep cycles, and high-stress environments may contribute to risks associated with cardiovascular disease (22). Furthermore, previous research in a study of nearly 500 career firefighters found that approximately 80% of the firefighters evaluated were classified as overweight (body mass index [BMI] > 25 kg·m^{-2}) and 34% were classified as obese (BMI >30 kg·m^{-2}), with a mean BMI of 28.6 ± 4.6 kg·m^{-2} (18), which further increases the risk for cardiovascular disease. Because of the high prevalence of obesity and heightened risk of cardiovascular disease, there has been increased attention directed toward improving overall fitness and health for firefighters (10,23). Annual physical fitness and health evaluations play an important role in this strategy as they provide insight into the current health status of these tactical athletes.

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Body composition assessment is a vital component of fitness evaluations, as it provides more detailed insight into a person's body stature, beyond that of body mass or BMI. Regular body composition evaluation can help health practitioners monitor changes in body composition–related training adaptations (i.e., fat loss, increase in lean body mass, etc.) and to best direct nutritional programming strategies to support fitness-related goals and optimize firefighter health. Two-compartment models are commonly used in field-based settings, in which select tissue compartments are characterized as fat mass and fat-free mass (FFM), whereas three-component models further characterize tissue components by measuring bone mineral content via dual-x-ray absorptiometry or total body water via impedance devices (11). Bioelectrical impedance analysis is a common methodological approach to estimating body composition using a doubly indirect body composition technique, relying on the transmission of an electrical current through body tissues, which measures the impedance, also referred to as resistivity and reactance, of the current through the body (11,14). These values are configured into customized algorithms to estimate body water and quantify tissue components, based upon body mass, to estimate body fat percent (14). The evolution of single- and multi-frequency analysis techniques, in conjunction with the accompanying algorithms, has substantially improved the accuracy and precision of this method of body composition assessment over time (21). This has subsequently led to advancements in multifrequency devices and predictive algorithms that may be available for commercial use or within athletic settings (14,21). These devices offer

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a noninvasive, easy-to-use, and financially affordable modality for body composition assessment.

Several consumer-grade devices have been investigated for their ability to accurately estimate body composition (3,4,9,14,15,20). Consumer-grade devices are often classified as bipolar or multipolar, depending upon the number of electrodes used. For example, instruments classified as bipolar involve the use of a current that travels from one end of a limb to another (e.g., hand-to-hand or foot-to-foot) (4). Select studies have reported handheld impedance devices can provide valid measures of body fat percentage (BF%) (12,20), whereas other studies have found them to significantly underestimate BF%, which subsequently overestimates FFM when compared with criterion measures (4,20). Similarly, previous research has produced conflicting findings regarding the accuracy of foot-to-foot impedance devices for estimating BF% in active populations (8,16,20,24). Impedance devices with multipolar leads are often more expensive, yet may provide more accurate estimates of fat mass as they use multiple frequencies, and multiple tactile electrodes often placed at more body sites (e.g., hand-to-foot) (2,4). Previous research assessing the validity of multifrequency impedance devices has reported a constant error (CE) of less than 1% and total error (TE) of ~3% in adults while providing an equivalent measure of BF% when compared with a 4-compartment criterion (20).

Although potentially sacrificing validity and reliability, consumer-grade impedance devices offer a convenient option for estimating body composition in field settings, which is advantageous for tactical populations that have limited time for evaluations while on duty. Therefore, the purpose of the current study was to examine the validity of consumer-grade impedance devices for the estimation of BF% in structural firefighters. It was hypothesized that multifrequency bioelectrical impedance analysis (MFBIA) would be the only impedance device to provide an acceptable measure of BF% and FFM.

Methods

Experimental Approach to the Problem

Subjects completed a battery of body composition assessments during a single morning of testing, including using MFBIA hand-to-foot device, a single-frequency BIA foot scale (F2FBIA), and a single-frequency handheld BIA device (HHBIA). Before testing, subjects were asked to refrain from intense physical activity (>24 hours) and food intake (>8 hours). To help promote a euhydrated state for testing, subjects were instructed to consume 1 L of water during the 8-hour food abstention period before arriving to the laboratory for testing. Upon arrival, subjects were asked to void their bladder before testing. Body composition parameters were also assessed using dual X-ray absorptiometry (DXA). Cross-validation occurred for all modes of body composition assessment, using the DXA measures of BF% and FFM as criterion measures. Target outcomes included BF% and FFM. All testing was completed by the same research staff member.

Subjects

Twenty-eight career structural firefighters participated in the study (female, $n = 2$; male, $n = 26$ [mean ± *SD*] age: 38.2 ± 8.3 years; height: 180.2 ± 7.5 cm; body mass: 86.7 ± 20.8 kg; BMI: 25.8 ± 7.8 $kg \cdot m^{-2}$). All interested subjects were invited to attend an informational meeting in which the details of the study, subject involvement, benefits, risks, and projected outcomes were discussed. Subjects who were actively cleared for duty were eligible to participate in the study. Exclusion criteria included women who were pregnant or breastfeeding. The study was conducted according to the Declaration of Helsinki guidelines and procedures were approved by the Mayo Clinic Institutional Review Board for the use of human subjects in research (IRB # 21–003343, approved on October 13, 2021).

Procedures

Anthropometrics. Body mass and height were assessed using a self-calibrating physician's scale and stadiometer to the nearest 0.1 kg and 0.5 cm, respectively.

Bioelectrical Impedance. Body composition was assessed using a MFBIA hand-to-foot device (H20N Scale, InBody Inc., Cerritos, CA); a single-frequency foot-to-foot BIA scale ([F2FBIA] Tanita BF-679W, IL), and a single-frequency handheld BIA device (HHBIA; Omron HBF-306) using manufacturer guidelines. Subjects wore light, loose-fitting clothing and removed shoes and socks during measurement. For the MFBIA device, height was entered before measurement. Subjects were instructed to stand upright, with arms extended, while holding the handle, with thumbs firmly pressed against the electrodes. Subjects were instructed not to let their thighs touch either, while the soles of their feet were firmly placed on the electrodes, without their feet touching. The MFBIA uses an 8-point tactile electrode system, using 10 impedance measurements taken by 2 frequencies at 20 and 100 kHz, with an average testing time of ~8 seconds. For the HHBIA device, data regarding height, sex, age, and weight were entered before measurement. Subjects were instructed to place both hands on the device while holding the grip electrodes, with the palm of the hand on the top and bottom electrodes. Subjects were then instructed to hold their arms fully extended at a 90-degree angle. For the F2FBIA device, height and sex were entered into the device and the "athlete" mode was selected before measurement.

Criterion Method. Criterion measures of BF% and FFM were determined using dual-x-ray absorptiometry (iDXA) (Lunar Prodigy, GE Healthcare) and the enCORE version 15.00 software. The DXA was calibrated daily before use, using the manufacturer-provided calibration block. Subjects were asked to remove any metal and metal-containing clothing before measurement and instructed to lie in a supine position, with their arms resting at their sides with palms in a prone position. All scans were completed by the same member of the research staff. Test-to-test reliability data from 11 adult subjects in our laboratory have yielded 95% confidence interval (CI) intraclass correlation coefficient values of 0.999–1.00 for FFM, 0.999–1.00 for lean body mass, 0.999–1.00 for fat mass, and 0.998–1.00 for BF%. The National Health and Nutrition Examination Survey body composition correction factor was not used.

Statistical Analyses

To determine which impedance methods (MFBIA, F2FBIA, and HHBIA) differed from the criterion method (iDXA), a 1-way analysis of variance test with repeated measures was performed, with the body composition assessment method specified as a within-subjects factor. Significant effects were followed up with pairwise *t*-tests, with the criterion DXA specified as the reference group and using the Holm adjustment for multiple comparisons.

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Bland-Altman analysis (1) with linear regression was performed to identify proportional bias, and the 95% limits of agreement were calculated to indicate individual-level error. Constant error (mean difference) between the criterion and BIA device was calculated. Correlations between the criterion method and BIA devices were established using Pearson's r correlation coefficient. The following criteria were used for interpreting correlation coefficients: very weak, <0.20; weak, 0.20–0.39; moderate, 0.40–0.59; strong, 0.60–0.79; and very strong, >0.80 (5). The standard error of the estimate (*SEE*) was estimated via regression procedures. Root-mean squared error, also known as TE, was calculated as follows:

$$\sqrt{\sum \frac{(BF\%_{estimated} - BF\%_{criterion})^2}{n}}$$

The TE is interpreted as the average deviation of individual values from the line of identity between the reference method and each device, whereas the *SEE* represents the deviation of individual data points around the line of best fit for the reference method and each device (20). Total error percentage (%) was also calculated by diving the TE by the mean of the criterion value for each respective device. The same statistical analysis procedures were performed for FFM. All data were analyzed using the IBM SPSS Statistics for Windows (Version 26.0; IBM Corp. Armonk, NY), and statistical significance was accepted at $p < 0.05$.

Results

Body fat % values produced by MFBIA, F2FBIA, and HHBIA were strongly associated ($p < 0.001$) with criterion BF% measures, with correlation coefficients of $r = 0.913$, $r = 0.695$, and $r = 0.876$, respectively. However, all BIA devices significantly underestimated BF% when compared with the criterion measure (Table 1). Constant error ranged between 4 and 5.5% across all BIA devices. Linear regression indicated the MFBIA had the closest agreement to the DXA criterion for BF% estimation ($R^2 = 0.834$, TE = 5.1, TE% = 20.1%), whereas the F2FBIA yielded the highest TE ($R^2 = 0.483$, TE = 7.4, TE% = 29.3%). Bland-Altman plots for BF% are presented in Figure 1. Bland-Altman analysis indicated that proportional bias was present for the F2FBIA device.

All BIA devices significantly overestimated FFM when compared with the criterion measure (Table 2). Constant error ranged between 3.4 and 4.7 kg across all BIA devices. FFM values produced by MFBIA, F2FBIA, and HHBIA were strongly associated ($p < 0.001$) with criterion FFM measures, as evidence by correlation coefficients of $r = 0.948$, $r = 0.850$, and $r = 0.905$, respectively. Linear regression indicated the MFBIA had the closest agreement to the DXA criterion for FFM estimation ($R^2 = 0.899$, TE = 4.6, TE% = 6.8%), whereas the F2FBIA yielded the highest TE ($R^2 = 0.722$, TE = 6.9, TE% = 10.1%). Bland-Altman plots for FFM are presented in Figure 2. Proportional bias was observed for all devices.

Table 1

Body fat percentage values and constant error across each device.*†

Device	Body fat (%)	CE	95% CI	*p*	*SEE*	R^2	TE	TE%
DXA	25.3 ± 7.4							
MFBIA	21.3 ± 5.2	4.05 ± 2.9	2.9–5.2	<0.001	3.01	0.834	5.1	20.1
F2FBIA	19.8 ± 5.2	5.46 ± 5.2	3.4–7.5	<0.001	5.32	0.483	7.4	29.3
HHBIA	20.9 ± 6.5	4.39 ± 3.6	2.9–5.8	<0.001	3.63	0.768	5.6	22.2

*CE = constant error; *SEE* = standard error of estimate; TE = total error; TE% = total error percent.
†Data presented as mean ± *SD*.

Discussion

The primary aim of the current study was to evaluate the validity of 3 consumer-grade BIA devices for the assessment of BF % in structural firefighters. Main findings from the current study indicate that despite strong associations between the BIA devices and the criterion measure, the BIA devices underestimate BF%, which resulted in an overestimation of FFM. In addition, proportional bias was observed in which BF% was overestimated at lower values and underestimated at higher values.

Although cost-efficient, portable, and easy to use, HHBIA devices may exhibit poor validity, despite being an appealing option for body composition assessment in field-based settings. Previous research has produced mixed findings regarding the validity of HHBIA devices (4,12,20,25,26). In the current study, when compared with the DXA, the HHBIA device resulted in a CE of 4.4% when estimating BF%, which is in alignment with previous findings from the study by Esco et al. (4) who reported a discrepancy in BF% values produced by a HHBIA device and those determined by DXA (DXA = 27.6 ± 5.3% vs. HHBIA = 22.5 ± 3.5%, $p < 0.01$), resulting in a mean difference of −5.11 ± 3.62% (95% CI of −6.26 to −3.95%, $p < 0.01$) and a correlation coefficient of $r = 0.74$. The authors (4) concluded that HHBIA devices may not be a valid alternative for BF% estimates, particularly in active women as they may underestimate BF%, as large limits of agreement between DXA and HHBIA measures of BF% were present for all subjects (−2.3 and 12.3%). By contrast, Lintsi et al. (12) observed a CE of only 0.2 ± 3.0% compared with a DXA-obtained BF% when estimating BF % using a HHBIA. However, the subjects in the study by Lintsi et al. (12) were younger (age 17–18 years vs. age 38 years in the current study), lighter (~74 vs. ~87 kg in the current study), and leaner (BF%: 14 vs. 25% in the current study) than the subjects in the current study, which may influence the validity of the device. When using the same HHBIA device as the one in the current study, Siedler et al. (20) reported a CE of −3.5% (95% CI: −4.4 to −2.5) compared with a 4C criterion, which was classified as "fairly good" by the authors, as further determined by equivalence testing and 90% confidence limits (−4.3 to −2.7). These discrepancies regarding the validity of HHBIA devices have been observed across multiple age groups and sex, even for the same device (26), indicating that HHBIA may not a valid tool for the measurement of BF% at an individual level because of the discrepancies observed and large ranges for limits of agreement. As such, HHBIA devices may only be suitable for group-level estimates of BF%.

Similar to HHBIA devices, F2FBIA devices offer a convenient, noninvasive, and low-cost option for body composition assessment. These devices resemble a consumer-grade scale, and often have bipolar or quadripolar electrodes for current transmission. More favorable outcomes have been reported regarding the validity of F2FBIA devices, compared with HHBIA devices. Nickerson et al. (16) reported no difference in BF% measurements between a F2FBIA and 3C criterion ($p = 0.14$), with a standard error of estimate of 2.79% and TE of 2.95% with 95% limits of agreement of ±5.67%, of which the authors considered acceptable for the determination of BF% in an active population.

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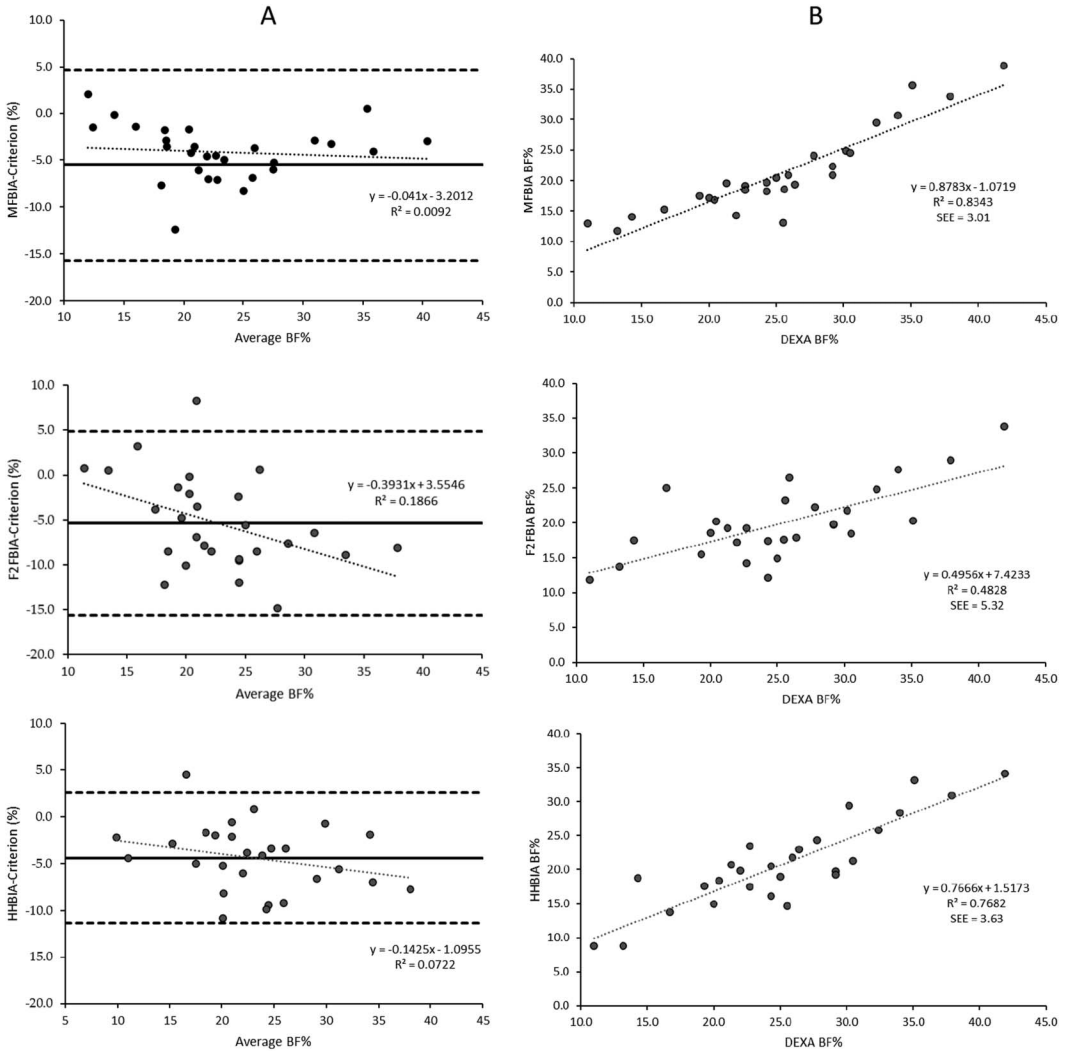

Figure 1. A) Bland-Altman analysis of body fat percentage values from BIA devices in firefighters. Horizontal dashed lines represent the 95% limits of agreement (i.e., 1.96 times the *SD* of the difference between methods), and the solid horizontal line represents the mean difference between methods. The diagonal line indicates the linear relationship between the difference between methods (*y*) and the average of the methods (*x*). A slope significantly different from zero indicates proportional bias. B) Each BIA device was compared with the criterion (DXA). Pearson's correlation coefficient squared, and standard error of the estimate (*SEE*) are displayed.

Ritchie et al. (19) reported minimal differences and a strong correlation between a F2FBIA device and MFBIA for BF% measurements (F2FBIA: 36.6 ± 9.7%; MFBIA: 35.5 ± 7.5%; $p = 0.13$; $r = 0.84$). Loenneke et al. (13) also reported an acceptable CE value of 0.6% when a F2FBIA device was compared with DXA for BF% measurement. However, while the F2FBIA device provided a valid estimate of BF%, it still resulted in a TE of 4.9%, which was classified as unacceptable according to the evaluation criteria used by the authors.

When using the same MFBIA device as the one in the current study, Siedler et al. (20) reported a CE of −0.1% (95% CI: −0.9 to 0.7) compared with a 4C criterion, which was classified as "very good" by the authors, as determined by equivalence testing and 90% confidence limits for two 1-sided *t*-tests (−0.8, 0.6). However, the CE of 4.05 from the MFBIA in the current study was higher than that reported by Siedler et al. (20). Dolezal et al. (2) reported minimal differences in measures of BF% using commercially available impedance devices (quadripolar, BIA4

Table 2

Fat-free mass values and constant error across each device.*†

Device	Fat-free mass (kg)	CE	95% CI	*p*	*SEE*	R^2	TE	TE%
DXA	68.1 ± 7.9							
MFBIA	71.5 ± 8.9	−3.4 ± 3.2	−4.6 to −2.1	<0.001	2.97	0.899	4.6	6.8
F2FBIA	72.8 ± 9.7	−4.7 ± 5.1	−6.7 to −2.7	<0.001	4.25	0.722	6.9	10.1
HHBIA	71.8 ± 8.2	−3.7 ± 3.5	−5.0 to −2.3	<0.001	3.43	0.819	5.0	7.4

*CE = constant error; TE = total error; TE% = total error percent.
†Data presented as mean ± *SD*.

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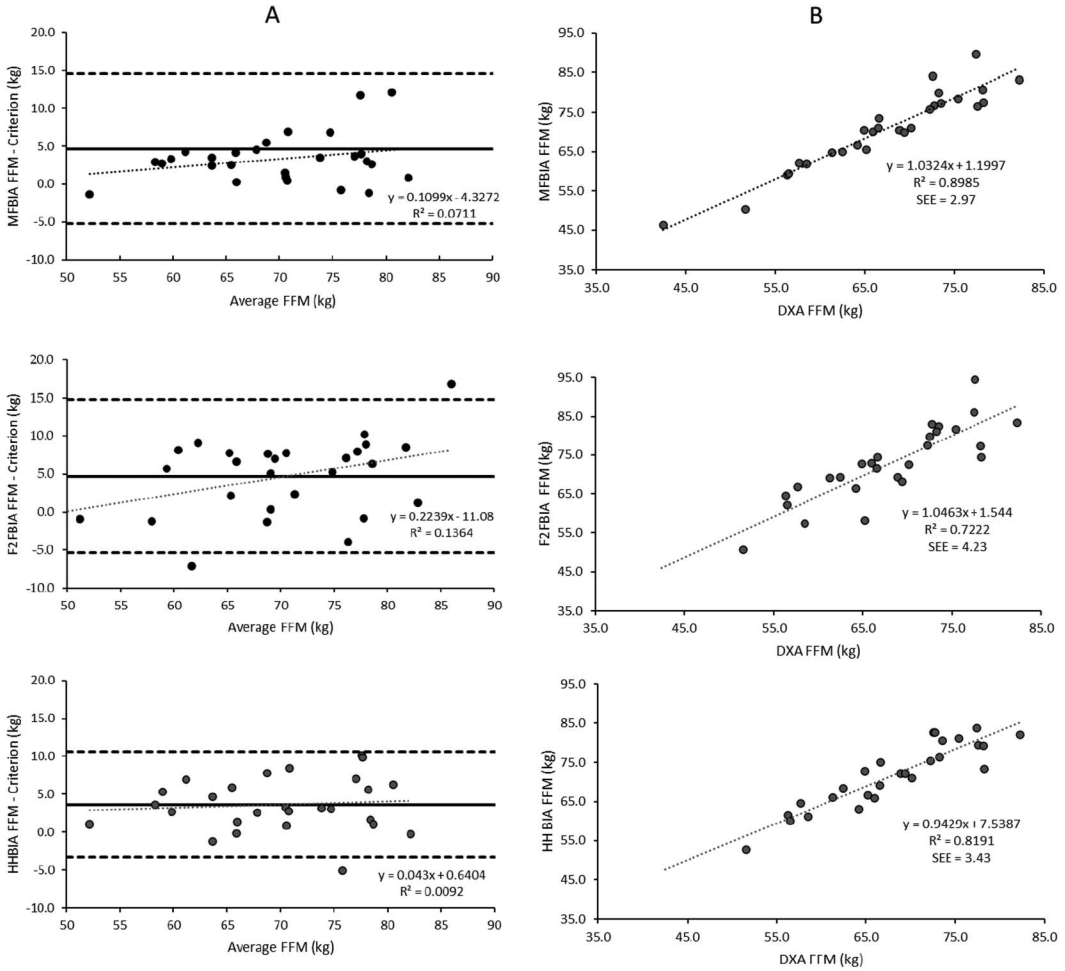

Figure 2. A) Bland-Altman analysis of fat-free mass values from BIA devices in firefighters. Horizontal dashed lines represent the 95% limits of agreement (i.e., 1.96 times the *SD* of the difference between methods), and the solid horizontal line represents the mean difference between methods. The diagonal line indicates the linear relationship between the difference between methods (*y*) and the average of the methods (*x*). A slope significantly different from 0 indicates proportional bias. B) Each BIA device was compared with the criterion (DXA). Pearson's correlation coefficient squared, and standard error of the estimate (*SEE*) are displayed.

and octapolar, BIA8). On average, differences of (mean ± *SD*) 0.14 ± 0.04 for BIA8 ($p = 0.80$) and 1.77 ± 0.54 for BIA4 ($p = 0.76$) in BF% were reported between for 2 BIA devices, when compared with a DXA criterion, which equated to mean errors of 1.6 and 4.6% for BIA8 and BIA4 vs. DXA, respectively.

Similar findings have been reported when evaluating FFM estimated from impedance devices. Esco et al. (4) found that HHBIA devices underestimated BF%, which subsequently overestimated measures of FFM (DXA = 47.2 ± 4.5 kg, BIA = 50.6 ± 4.6 kg, $p < 0.01$) compared with a DXA criterion in active women. Nickerson et al. (16) initially reported that F2FBIA could provide acceptable measures of BF% when compared with DXA. Similarly, the authors also observed FFM values derived from F2FBIA were not statistically different from those determined by a 3C criterion ($p = 0.28$), with a standard error of estimate of 2.64 kg, TE of 2.64 kg, and 95% limits of agreement of ±5.15 kg, which the authors considered acceptable for the determination of FFM. In the current study, estimates of FFM from the MFBIA, F2FBIA, and HHBIA were strongly associated ($p < 0.001$) with criterion FFM measures. However, all BIA devices significantly overestimated FFM when compared with the criterion measure (Table 2). Therefore, the consumer-grade BIA devices evaluated in the current study may not provide valid estimates of FFM for firefighters.

It is challenging to identify the primary reason for such variability across studies regarding the validity of BIA devices. There may be inherent anthropometric, anatomical, or body water discrepancies across certain populations or sex that influence the accuracy and precision of these devices in certain circumstances. In addition, the selection of specific settings on certain impedance devices may affect the accuracy of the BF% estimate (24). Swartz et al. (24) reported that the choice of "activity" mode setting on a F2FBIA device significantly altered the accuracy of BF% estimation. However, when the selection of activity mode matched the self-reported activity levels of the subjects, there was no difference between BF% estimated by the F2FBIA compared with an underwater weighing criterion method. With consumer-grade BIA devices, it is often not possible to obtain the raw impedance values or identify the prediction equation used for estimating BF %, which precludes the ability to identify some of the contributing factors influencing the agreement of each device.

This study is not without limitations. The lack of a criterion measure for total body water, such as that from deuterium, precludes the ability of a more comprehensive model of body

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composition to use as a criterion. Moreover, the use of four-compartment model would have provided a more robust model to use as a criterion measure. However, previous research has reported minimal differences in measures of BF% between three- and four-compartment models (7,17,27). The lack of urine-specific gravity assessment before testing prohibited confirmation of hydration status and is another limitation of the current study. However, pretesting guidelines regarding dietary intake, fluid intake, and activity were provided to subjects before testing.

Practical Applications

Despite strong associations between the BIA devices included in the current study and the criterion measure, the BIA devices underestimated BF%, which resulted in an overestimation of FFM. In addition, proportional bias was observed in which BF % was overestimated at lower values and underestimated at higher values. The consumer-grade BIA devices evaluated in the current study may only be appliable for estimating BF% and FFM at the group level for general descriptive purposes among tactical populations and may have limited application for assessing body composition at the individual level, particularly when a precise measure of BF% or FFM is required.

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